🧑‍🏫 **Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.**

UNIT 1 · STANDARD 6.NOS.4

Least Common Multiple

Lesson 1-3 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

My Notes Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can find the least common multiple (LCM) of two numbers by listing or comparing their multiples.
Language Objective: I can explain how I found the LCM using the words multiple, common multiple, skip counting, and least common multiple.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Multiple | **Least common multiple** | **Common multiple** | **Skip counting** | **Prime factorization**

👆 *Tap any word to see what it means and a picture.*

Level 3: try the blanks from memory first, then check the bank.

1. The product of a number and any whole number is a ________ of that number.

2. ________ — The smallest number that two or more numbers both go into.

3. A multiple that two or more numbers share is a ________ multiple.

4. Counting forward by a number other than 1, like 6, 12, 18, is called ________.

5 ______ — Writing a number as prime numbers multiplied together. It helps you find the LCM.

Write About the Math **Guided ESOL writing**

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

You skip counted the multiples of 4 and 6. How did you find the least common multiple from your lists?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

- [ ] **Multiple** — What you get when you multiply a number by 1, 2, 3, and so on.
  Múltiplo — Lo que obtienes al multiplicar un número por 1, 2, 3, y así.

- [ ] **Least common multiple** — The smallest number that two or more numbers both go into.
  Mínimo común múltiplo — El número más pequeño al que llegan dos o más números al multiplicar.

- [ ] **Common multiple** — A number that two or more numbers both go into.
  Múltiplo común — Un número al que llegan dos o más números al multiplicar.

- [ ] **Skip counting** — Counting by a number, like 2, 4, 6, to list its multiples.
  Conteo salteado — Contar de un número en un número, como 2, 4, 6, para listar sus múltiplos.

- [ ] **Prime factorization** — Writing a number as prime numbers multiplied together. It helps you find the LCM.
  Factorización prima — Escribir un número como números primos multiplicados. Ayuda a encontrar el mcm.

Say it first: Say your idea to a partner or quietly to yourself before you write.
Di tu idea a un compañero o en voz baja antes de escribir.

The multiples of 4 are ____ and the multiples of 6 are ____.
Los múltiplos de 4 son ____ y los múltiplos de 6 son ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: The next shared check is a common multiple of 4 and 6, not their sum.

Evidence: Students reason that each astronaut repeats on multiples of their own number, so they only overlap at a common multiple, and adding 4 + 6 does not land on both schedules.

Reasoning: This evidence connects the definition of multiple to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The multiples of 4 are ____ and the multiples of 6 are ____.**
  Los múltiplos de 4 son ____ y los múltiplos de 6 son ____.
- **The least common multiple is ____ because it is the ____ shared multiple.**
  El mínimo común múltiplo es ____ porque es el múltiplo compartido más ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **I know because ____.**
  Lo sé porque ____.
- **So ____.**
  Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
  Mi afirmación es ____.
- **My math evidence is ____.**
  Mi evidencia matemática es ____.
- **This proves ____ because ____.**
  Esto demuestra ____ porque ____.

4. Check Your Explanation

- [ ] I answered the question.
  Respondí la pregunta.
- [ ] I used math evidence: numbers, an equation, a model, or a comparison.
  Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- [ ] I explained how the evidence proves my answer.
  Expliqué cómo la evidencia demuestra mi respuesta.
- [ ] I used at least two math words.
  Usé por lo menos dos palabras matemáticas.
- [ ] I reread complete sentences and punctuation.
  Volví a leer mis oraciones completas y la puntuación.

Writing Guide — Teacher Copy

The support level changes the amount of language scaffolding, not the mathematical expectation. Look for the same five features in every response.

- I answered the question.
- I used math evidence: numbers, an equation, a model, or a comparison.
- I explained how the evidence proves my answer.
- I used at least two math words.
- I reread complete sentences and punctuation.

Answer Key & Teacher Guide

1. **Notes 1:** Multiple
2. **Notes 2:** Least common multiple
3. **Notes 3:** Common multiple
4. **Notes 4:** Skip counting
5. **Notes 5:** Prime factorization
6. **Watch for this mistake:** *A common mistake in Least Common Multiple is multiplying the two numbers together instead of listing multiples to find the smallest shared one. For example, for 4 and 6, a student calculates 4 × 6 = 24 and calls that the LCM, but the actual least common multiple is 12 (multiples of 4: 4, 8, 12; multiples of 6: 6, 12 — 12 is the first number that appears in both lists). This shortcut only works when the two numbers share no common factors (like 4 and 9, where the LCM really is 36 = 4 × 9); for numbers like 4 and 6 that share a factor of 2, multiplying overshoots the true LCM. Always list multiples of both numbers and pick the first one that appears in both lists.*
7. **Try It 1:** A. 12 — *Skip count: multiples of 4 are 4, 8, 12; multiples of 6 are 6, 12. The first shared multiple is LCM(4, 6) = 12 minutes.*
8. **Try It 2:** A. 24 — *LCM(8, 12) = 24. Multiples of 8: 8, 16, 24; multiples of 12: 12, 24. The first shared multiple is 24 minutes.*